Project Proposal:
Comparing CNN and Transformer Representations for Low-Data Land-Use Classification & Data Augmentation

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Project Summary

Many real-world image classification tasks have limited labeled data because annotation requires time and resources. Pretrained models and data augmentation are used to address this limitation, but their benefits may vary across models and augmentation methods.

This project will compare four pretrained models—ResNet-18, ConvNeXt-Tiny, ViT-B/16, and DINOv2 ViT-S/14—using one, five, and ten labeled images per class for satellite image classification. Using frozen pretrained backbones and a common downstream classifier, we will examine how performance and augmentation benefits change with labeled-data availability and identify effective model–augmentation pairs.

The findings could guide model and augmentation selection for satellite imagery with limited labels, particularly in remote regions.

Approach

Each model will be evaluated to answer the following question:

How do CNN- and Transformer-based visual representations compare for land-use classification when only a small number of labeled examples are available?

- ResNet-18: A standard CNN baseline.
- ConvNeXt-Tiny: A modern CNN compared with the ResNet-18 baseline.
- ViT-B/16: A standard Vision Transformer compared with the CNN models.
- DINOv2 ViT-S/14: A Vision Transformer pretrained through self-supervised learning, used to evaluate the transfer performance of this representation.

We will use ImageNet-1K supervised checkpoints for ResNet-18, ConvNeXt-Tiny, and ViT-B/16, and the official dinov2_vits14 checkpoint for DINOv2. Exact checkpoint identifiers and software versions will be recorded in the shared experimental configuration.

We will focus on the following questions:

- How well do conventional CNN representations transfer from natural-image pretraining to satellite imagery?
- Do modern CNN architectures improve upon a standard ResNet baseline?
- Do the selected Vision Transformers provide better representations than the selected CNNs under limited labeled data?
- Does a large-scale self-supervised representation such as DINOv2 provide additional benefits in the few-shot setting?
- How does the relative performance of the models change from 1-shot to 5-shot to 10-shot classification?

The following data augmentation methods will be applied to answer the following questions:

Does data augmentation help compensate for a limited number of labeled images? Do different models benefit from different augmentation methods?

- Geometric Transformations
    - Flipping
    - Rotation
    - Translation
    - Scaling
- Color and Pixel Modifications
    - Brightness and contrast changes
    - Noise injection
    - Blurring
- Information Dropping
    - Random erasing
    - Cutout, as an optional comparison

We will focus on the following questions:

- Which augmentation methods improve classification performance compared with no augmentation?
- How much can augmentation reduce the performance gap between fewer and more labeled original images?
- Does increasing the number of augmented views provide additional benefits?
- Do augmentation methods produce different performance improvements across ResNet-18, ConvNeXt-Tiny, ViT-B/16, and DINOv2?
- Are augmentation benefits larger in the 1-shot setting than in the 5-shot or 10-shot settings?

Resources and Related Work

EuroSAT: A Novel Dataset and Deep Learning Benchmark for Land Use and Land Cover Classification — Helber et al. (2019) https://arxiv.org/abs/1709.00029

- Provides the EuroSAT dataset description and original land-use classification benchmarks.

A Broader Study of Cross-Domain Few-Shot Learning — Guo et al. (2020) https://arxiv.org/abs/1912.07200

- Provides an evaluation framework for transferring natural-image representations to satellite imagery with limited labels.

Do Vision Transformers See Like Convolutional Neural Networks? — Raghu et al. (2021) https://arxiv.org/abs/2108.08810

- Provides background on differences between CNN and ViT representations.

DINOv2: Learning Robust Visual Features without Supervision — Oquab et al. (2023) https://arxiv.org/abs/2304.07193

- Supports evaluating large-scale self-supervised representations using DINOv2 as a frozen feature extractor.

RandAugment: Practical Automated Data Augmentation with a Reduced Search Space — Cubuk et al. (2020) https://arxiv.org/abs/1909.13719

- Provides background for defining augmentation policies and controlling transformation strength.

Datasets

Dataset link: EuroSAT RGB Dataset

Official dataset repository: EuroSAT

We will use EuroSAT RGB, which contains approximately 27,000 Sentinel-2 satellite image tiles.

- Image size: 64 × 64 pixels
- RGB imagery
- 10 land-use/land-cover classes
- Input preprocessing: Resize to 224 × 224 using bilinear interpolation and apply checkpoint-appropriate normalization.

All 10 classes will be retained for a 10-way classification task. We propose a stratified split of 60% for the support-sampling pool, 20% for development, and 20% for held-out query/test evaluation. Duplicates will be checked before fixing the split.

For each evaluation trial, we will sample:

- 1-shot: 1 original image per class, or 10 images in total
- 5-shot: 5 original images per class, or 50 images in total
- 10-shot: 10 original images per class, or 100 images in total

Support images will be used to fit the classifier, while held-out query images will be used for evaluation. Development labels used for selecting settings will be reported separately; the shot budget refers to classifier fitting in each trial.

Augmented views will not count as additional shots. All variants of an original image will remain in the same partition.

Evaluation Plan

All four models will follow the same evaluation procedure as closely as possible.

Satellite image → optional support-image augmentation → pretrained model → visual embedding → common few-shot classifier → prediction

The downstream classifier will be standardized across models. We will use an L2-regularized multinomial logistic regression classifier trained on L2-normalized embeddings.

The pretrained classification heads will be removed. CNN embeddings will be extracted from their pooled features before the classification head, and Transformer embeddings will be extracted from their final normalized class-token representations. Exact extraction points will be documented for each model.

We will keep the pretrained backbones frozen in evaluation mode and train only the downstream classifiers, using the same classifier type to compare existing representations under consistent conditions. We will test whether support-image augmentation improves classifier generalization when original labeled images are limited.

We will also use consistent:

- Data partitions
- Input size and resizing
- Few-shot support sampling
- Augmentation policies and random seeds
- Classifier selection procedures
- Evaluation metrics
- 1/5/10-shot settings

The same support and query images will be used across all models and augmentation conditions. Where possible, the 1-shot support images will be included in the corresponding 5-shot and 10-shot support sets.

If time and computational resources permit, fine-tuning will be considered as an optional extension. It will be reported separately from the frozen-backbone experiments.

Few-Shot Performance

The primary experiment will compare accuracy under:

- 1-shot
- 5-shot
- 10-shot

CNN vs Transformer

- ResNet-18 vs ConvNeXt-Tiny
- ViT-B/16 vs DINOv2 ViT-S/14
- The selected CNN models vs the selected Transformer models

The primary comparison will focus on whether the selected Transformer-based representations show an advantage under severe label scarcity.

ViT-B/16 and DINOv2 differ in model size, pretraining data, and learning approach. Their comparison will evaluate the usefulness of the available pretrained representations, but it will not isolate the effect of self-supervised learning alone.

Data Augmentation

Each augmentation method will first be evaluated separately using the same original support images. The proposed main augmentation condition will contain the original image plus four augmented views per original support image. The same generated views will be used across models.

We will compare:

- Original support images without augmentation
- Original support images with one augmentation method

- Repeated copies of the original support images, matched to the augmented-view count

To examine whether augmentation compensates for limited labeled data, we will compare:

- 1-shot without augmentation vs 1-shot with augmentation
- 5-shot without augmentation vs 5-shot with augmentation
- 10-shot without augmentation vs 10-shot with augmentation
- 1-shot with four augmented views vs 5-shot without augmentation
- 1-shot with nine augmented views vs 10-shot without augmentation, for selected methods

Augmentation will be applied only to support images. Development and test query images will receive deterministic preprocessing without random augmentation or test-time augmentation.

Model-Specific Augmentation Effects

For each model and shot setting, we will report both the final accuracy and the accuracy improvement over the corresponding no-augmentation baseline.

Accuracy improvement will be reported in percentage points. We will compare paired improvements across trials to examine whether the same augmentation produces different benefits for different representations.

A model's preferred augmentation will be defined as the method selected using development performance. Its benefit will then be checked on the held-out test data. We will also report the full set of prespecified augmentation comparisons so that small or inconsistent improvements remain visible.

Class-Level Performance

We will examine Macro-F1, class-level recall, and confusion matrices to identify whether certain types of satellite imagery are more difficult for particular models.

We will also examine whether an augmentation helps some land-use classes while reducing performance on others.

Computational Cost

We will record basic computational characteristics such as model size, feature-extraction time, classifier-fitting time, and memory use where practical.

Original and augmented embeddings will be cached separately for each model and augmentation configuration. Hardware, software versions, and timing conditions will be documented.

Risks & Solution

- Differences in pretraining: Document checkpoint and model differences when interpreting results.
- Sampling variability: Repeat evaluations using identical support sets across conditions.

- Augmentation strength: Use development data to avoid removing class-defining information.
- Computational constraints: Cache embeddings and prioritize individual augmentation experiments.
- Implementation inconsistencies: Share preprocessing, feature-extraction, and evaluation code.
- Data leakage: Split originals before augmentation and keep development and test data separate.
- Spatial similarity: Use geographic grouping where reliable metadata permits; otherwise acknowledge limitations in geographic generalization.
- Multiple comparisons: Select augmentation methods using development data and report inconsistent or negative effects.

Ethical Implications

We will follow EuroSAT's usage and attribution requirements and report performance variability, development-label usage, and limitations transparently. Augmented views will not be presented as independently labeled images. Since classification errors could affect environmental monitoring and land-use planning, practical applications would require further validation across regions and imaging conditions.